

Praneel Magapu

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SUMMARY

Computer Science senior at NC State with 5+ years of programming experience in Python, Java, and C++. Built distributed backend services, real-time data pipelines, computational geometry systems, and AI/ML applications through industry internships, university research, and engineering projects. U.S. citizen.

EDUCATION

North Carolina State University

Aug 2023 – May 2027

Bachelor of Science in Computer Science, GPA: 3.5/4.0

Raleigh, NC

Relevant Coursework: Data Structures, Operating Systems, Software Engineering, Artificial Intelligence

TECHNICAL SKILLS

Languages: C++, Python, Java, C, JavaScript, TypeScript, Bash

Systems / Graphics: OpenCV, Eigen, Open3D, NumPy, CMake, Trimesh

Frameworks / Libraries: Flask, FastAPI, React, Tailwind CSS, Pydantic, NLTK

Tools / Platforms: Git, Docker, AWS (Elastic Beanstalk, CLI), Supabase, Firebase, Jenkins, GitHub Actions

AI & Data: LLM integration, RAG systems, Multimodal AI (TwelveLabs), NLP, MongoDB, SQL

EXPERIENCE

Align Technology

May – Aug 2026

Software Engineering Intern

Raleigh, NC

- Redesigned support-structure generation in a large C++ pipeline for directly 3D-printed dental devices, reducing manual finishing time per device and material use while improving surface quality on the final part.
- Implemented mesh-processing algorithms that compute valid placement regions directly from part meshes, generalizing across device geometries without per-case tuning.
- Applied OpenCV image-processing techniques to rasterized geometry to compute pixel-level support contact footprints, optimizing support-part interfaces for clean detachment after printing.

DevDynamics.ai

May – Aug 2025

Software Engineering Intern

Remote

- Built a hybrid regex and LLM classifier over Google Workspace APIs that resolved thread state across 100+ active email threads, escalating only low-confidence cases to the model to hold down inference cost.
- Shipped an AI-powered Gmail auto-labeler on Flask, AWS, and Supabase that replaced manual inbox triage for the internal team.

Liquid Rocketry Lab, NC State

Sep 2024 – May 2025

Data Engineer

Raleigh, NC

- Built the real-time telemetry pipeline for live rocket engine tests in Python and MongoDB across containerized microservices, capturing abort and breach events at sub-second latency.
- Designed an observable fault-detection pattern driving automated abort detection and breach logging, hardening safety response under high-frequency sensor ingestion.

North Carolina State University

May 2024 – Dec 2025

Research Assistant

Raleigh, NC

- Game2Learn Lab (Dr. Tiffany Barnes): built the production RAG system behind MerryQuery, a generative AI teaching assistant, and drove full-stack integration through a live classroom pilot.
- Crowd Label Quality Control (Dr. Ed Gehringer): extended interval-analysis and iterative aggregation algorithms to flag unreliable annotators at scale, raising label integrity across large ML training sets.

PROJECTS

Mendacia | Python, Flask, React, TypeScript, TwelveLabs, LLM APIs, Tailwind CSS

Hack_NCState 2026

- Built a full-stack multimodal media forensics platform generating structured forensic reports from raw video, detecting propaganda and cross-modal inconsistencies using TwelveLabs scene intelligence and LLM reasoning.
- Designed a cross-modal consistency engine comparing narrative claims against visual scene data with timestamped reasoning, plus a defensive adapter layer enforcing a stable UI contract across volatile backend responses.

Mesh Alignment Engine | Python, Open3D, NumPy, Trimesh

Mar – May 2026

- Implemented an ICP-based surface mesh registration pipeline aligning 3D scans with sub-millimeter RMSE, applying a multi-resolution coarse-to-fine strategy to accelerate convergence on high-density meshes.
- Built a deviation heatmap visualization mapping per-point geometric divergence between aligned meshes.